PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	THE EFFECTS OF PREMATURITY ON LONG-TERM RENAL HEALTH: A SYSTEMATIC REVIEW
AUTHORS	Sangla, Ananya; Kandasamy, Yogavijayan

VERSION 1 – REVIEW

REVIEWER	Matthew Harer University of Wisconsin-Madison School of Medicine and Public Health, Pediatrics
REVIEW RETURNED	25-Jan-2021

GENERAL COMMENTS	In this systematic review, Ananya Sangla and Yogavijayan Kindasamy tackle a very important and timely topic, the long term kidney outcomes of preterm birth. There are a few important articles that I think are missing and should be included. Similarly, I think they should have statistical help to develop forest plots for GFR and hypertension which would greatly improve this manuscript. Ultimately, I disagree with some of their conclusions and suggest they be changes to match their findings. I don't think any absolute conclusions can be made from the currently available literature as the studies are primarily small and single center. Much of this data will be hypothesis generating for future larger more complete studies. My specific comments by section are presented below. KDIGO recently presented nomenclature guidelines. The full KDIGO nomenclature guideline is published in Kidney International - https://www.kidney-international.org/article/S0085-2538(20)30233-7/pdf - followed by accompanying editorials in other kidney and paediatrics journals. In brief: Where reasonable, please replace 'renal' with 'kidney' eg in 'renal replacement therapy'. Abstract: - The conclusion overstates the findings. A more accurate statement may go, "Small studies support kidney abnormalities in child born preterm." Introduction: - Throughout the paper the authors use prematurity and preterm birth interchangeably. The manuscript would be improved by picking on of the terms and sticking with it throughout. - I would remove the specific data on Australia (Page 3 lines 22-28, Page 4 lines 8-10) – your paper will be read worldwide and not just Australia. If it goes to a regional publisher, then this may be appropriate. - Paragraph 3 is unnecessary and not relevant to this publication and should be removed entirely. This data deserves to be in a book chapter and not a systematic review. - Good last two paragraphs of the introduction.

	Methods: - Relevant keywords: the list feels very short. I would have included things like proteinuria, albuminuria, hypertension, high blood pressure, reduced eGFR, decreased kidney function. - Page 7, line 49 – "…or having low birth weight." This was not one of your original goals, so why would they have been included? This could have added term infants who were low birth weight but not premature. - It appears you are missing some important publications on this topic: 1. Bruel A, Rozé JC, Quere MP, Flamant C, Boivin M, Roussey-Kesler G, Allain-Launay E. Renal outcome in children born preterm with neonatal acute renal failure: IRENEO-a prospective controlled study. Pediatr Nephrol. 2016 Dec;31(12):2365-2373. doi: 10.1007/s00467-016-3444-z. Epub 2016 Jun 22. PMID: 27335060. 2. Maqsood S, Fung N, Chowdhary V, Raina R, Mhanna MJ. Outcome of extremely low birth weight infants with a history of neonatal acute kidney injury. Pediatr Nephrol. 2017 Jun;32(6):1035-1043. doi: 10.1007/s00467-017-3582-y. Epub 2017 Feb 14. PMID: 28194575. 3. Harer MW, Pope CF, Conaway MR, Charlton JR. Follow-up of Acute kidney injury in Neonates during Childhood Years (FANCY): a prospective cohort study. Pediatr Nephrol. 2017 Jun;32(6):1067-1076. doi: 10.1007/s00467-017-3603-x. Epub 2017 Mar 3. PMID: 28255805. 4. Askenazi DJ, Heagerty PJ, Schmicker RH, Brophy P, Juul SE, Goldstein SL, Hingorani S; PENUT Trial Consortium. The Impact of Erythropoietin on Short and Long-term Kidney-Related Outcomes in Extremely Low Gestational Age Neonates. Results of a Multi-center Double-Blind Placebo-Controlled Randomized Clinical Trial. J Pediatr. 2021 Jan 20:S0022-3476(21)00065-2. doi: 10.1016/j.jpeds.2021.01.031. Epub ahead of print. PMID: 33484699. Results: - Page 9, line 42 "the smallest cohort size was 19…" In the text above, it states those with less than 20 were excluded. How was this article included? - Nephrocalcinosis – I have a really hard time as including this as an outcome for long term renal health of prematurity as your title implies. It is certainly an important short term outcomes in the first few years – but ultimately most nephrocalcinosis resolves after stopping diuretics for BPD and isn't something that fits into the definition of CKD. I would consider removing the papers about nephrocalcinosis only and removing this section or changing the title of your paper. - GFR – This entire section is difficult to read and unfocused. The paragraphs should be re-arranged to tell a story of the results that is logical and not be a list of what each study found. o Where is the forest plot for this outcome? - Blood pressure o Where is the forest plot for his outcome? - There should be a paragraph describing the outcomes of proteinuria or albuminuria in this population. Discussion: - Page 16 – line 20 "thus it is likely that in the short term, GFR

	is not significantly affected by prematurity". This conclusion does not reflect the data presented in the article. It should be modified. - Line 25, "this is likely the case…" I don't think we know why. The reduced nephron number is one theory, but there are many contributing factors. To make a guess like this is fine, but it should be framed as a hypothesis, not certainty. - Line 28-32 – this statement is out of place as your results didn't talk about CAD, obesity or metabolic disease, please remove both sentences. - Lines 37-55 – this is an incorrect conclusion of your data. In your results you point out 4 studies that did find differences. Perhaps these are the best studies and methodology and the others had poor methodologies and incorrect findings – or not the right comparison groups, or too few of patients. This paragraph needs to be re-done. - Page 17, nephrocalcinosis discussion – would remove this paragraph entirely. Conclusion - Line 42 – I would remove your age range and say childhood. - Line 45 – your results do not support this statement and it should be removed.

REVIEWER	Keia Sanderson UNC Kidney Center, Nephrology and hypertension
REVIEW RETURNED	19-Feb-2021

GENERAL COMMENTS	This is important work. I enjoyed reading and reviewing this. Would revise a few things as followed: 1. Conceptually you excluded two key papers with cohorts greater than 20 which observed long term renal outcomes after preterm birth. (South AM, Nixon PA, Chappell MC, et al. Renal function and blood pressure are altered in adolescents born preterm. Pediatr Nephrol. 2019;34(1):137-144. doi:10.1007/s00467-018-4050-z) (Sanderson KR, Chang E, Bjornstad E, et al. Albuminuria, Hypertension, and Reduced Kidney Volumes in Adolescents Born Extremely Premature. Front Pediatr. 2020;8:230. Published 2020 May 12. doi:10.3389/fped.2020.00230) 2. Following this point you state that you excluded cohorts where the n is less than 20 but in the second paragraph of the results, your very first sentence states that "Of the included studies, the smallest cohort size was 19.." this is contradictory to your exclusion criteria described in your methods and confusing to readers. 3. Also conceptually, you discuss renal dysfunction and hypertension in the intro as an explanation for your research question but nephrocalcinosis is under-developed from the very beginning. So it raises the question why did you evaluate nephrocalcinosis as an endpoint when it does not seem to be a part of the Brenner hypothesis or other prior research that you describe in the intro? --- INTRODUCTION 1. The introduction is too long. I would recommend removing the entire 3rd paragraph as it moves into neurocognitive and metabolic disease outcomes which are not the focus of this paper. 2. Page 6 Line 13. Would remove "become" and replace with "is" as it will make the reader stumble. "As the ex-preterm population is living longer....understanding the effects of prematurity IS imperative in anticipating the likely chronic health outcomes...." ---

	RESULTS 1. On page 11 line 24: Would correct as follows "Yael et al found that all 103 study participants with a history of very low birth weight and preterm birth had normal GFR values." Also you need to add at what age was this population studied as you did with other studies in this paragraph. 2. On page 11 line 37: Would be consistent with representation of numbers. "Outcomes were measured at approximately 7.5 years, 8 years, 8.5 years..." And throughout the text would be consistent with representation of numbers (with the exception of a number at the beginning of a sentence which should always be spelled out). Its simple but can make things appear sloppy. --- DISCUSSION 1. Ironically, this paragraph is started with a number 25 instead of the word "Twenty-five." Would start sentence with a number spelled out.

VERSION 1 – AUTHOR RESPONSE

Reply to reviewers' and editorial comments

Reviewer 1

In this systematic review, Ananya Sangla and Yogavijayan Kandasamy tackle a very important and timely topic, the long-term kidney outcomes of preterm birth. There are a few important articles that I think are missing and should be included.

Similarly, I think they should have statistical help to develop forest plots for GFR and hypertension which would greatly improve this manuscript.

Ultimately, I disagree with some of their conclusions and suggest they be changes to match their findings. I don't think any absolute conclusions can be made from the currently available literature as the studies are primarily small and single center. Much of this data will be hypothesis generating for future larger more complete studies.

Reply: We like to thank the reviewer for the comments and suggestion. We will address them in our resubmission.

My specific comments by section are presented below.

KDIGO recently presented nomenclature guidelines. The full KDIGO nomenclature guideline is published in Kidney International - https://www.kidney-international.org/article/S0085- 2538(20)30233-7/pdf - followed by accompanying editorials in other kidney and paediatrics journals. In brief: Where reasonable, please replace 'renal' with 'kidney' eg in 'renal replacement therapy'.

Reply: We have replaced ‘renal’ with ‘kidney’ wherever possible within the manuscript.

Abstract:

- The conclusion overstates the findings. A more accurate statement may go, “Small studies support kidney abnormalities in child born preterm.”

Reply: We have changed this statement to make it more in line with the study findings.

Introduction:

- Throughout the paper the authors use prematurity and preterm birth interchangeably. The manuscript would be improved by picking on of the terms and sticking with it throughout.

Reply: We have removed ‘preterm’ throughout the manuscript where appropriate and used ‘premature/prematurity’ instead.

- I would remove the specific data on Australia (Page 3 lines 22-28, Page 4 lines 8-10) – your paper will be read worldwide and not just Australia. If it goes to a regional publisher, then this may be appropriate.

Reply: The data specific to Australia has been removed.

- Paragraph 3 is unnecessary and not relevant to this publication and should be removed entirely. This data deserves to be in a book chapter and not a systematic review.
- Good last two paragraphs of the introduction.

Reply: We have removed paragraph 3 of the introduction

Methods:

- Relevant keywords: the list feels very short. I would have included things like proteinuria, albuminuria, hypertension, high blood pressure, reduced eGFR, decreased kidney function.

Reply: Additional terms have been added to the list of relevant keywords.

- Page 7, line 49 – "…or having low birth weight." This was not one of your original goals,

so why would they have been included? This could have added term infants who were

low birth weight but not premature.

Reply: We noticed that many studies investigated low birth weight and renal function as well. We decided to include these articles as long as the low-birth-weight cohorts were also premature. We have added a line about this in the last paragraph of our methods section (around Page 8 line 50).

- It appears you are missing some important publications on this topic:

1. Bruel A, Rozé JC, Quere MP, Flamant C, Boivin M, Roussey-Kesler G, Allain-Launay E. Renal outcome in children born preterm with neonatal acute renal failure: IRENEO-a prospective controlled study. Pediatr Nephrol. 2016 Dec;31(12):2365-2373. doi: 10.1007/s00467-016-3444-z. Epub 2016 Jun 22. PMID: 27335060.

2. Maqsood S, Fung N, Chowdhary V, Raina R, Mhanna MJ. Outcome of extremely low birth weight infants with a history of neonatal acute kidney injury. Pediatr Nephrol. 2017 Jun;32(6):1035-1043. doi: 10.1007/s00467-017-3582-y. Epub 2017 Feb 14. PMID: 28194575.

3. Harer MW, Pope CF, Conaway MR, Charlton JR. Follow-up of Acute kidney injury in Neonates during Childhood Years (FANCY): a prospective cohort study. Pediatr Nephrol. 2017 Jun;32(6):1067-1076. doi: 10.1007/s00467-017-3603-x. Epub 2017 Mar 3. PMID: 28255805.

4. Askenazi DJ, Heagerty PJ, Schmicker RH, Brophy P, Juul SE, Goldstein SL, Hingorani S; PENUT Trial Consortium. The Impact of Erythropoietin on Short and Long-term Kidney-Related Outcomes in Extremely Low Gestational Age Neonates. Results of a Multi-center Double-Blind Placebo-Controlled Randomized Clinical Trial. J Pediatr. 2021 Jan 20:S0022-3476(21)00065-2. doi:

10.1016/j.jpeds.2021.01.031. Epub ahead of print. PMID: 33484699.

Study	Population	AKI definition	Findings	Other
Abitbol et al. [84]	Born < 1000 g Age of F/U = 7.5 yrs AKI (20)	ARF (peak sCr > 2.0 for 48 h, UOP < 0.5 for 24 h)	– 55% with low GFR (sCr) – 25% with ESRD (four requiring transplant)	
Bruel et al. [86]	Born < 33 weeks Age of F/U = 6.5 yrs AKI (n = 25) No AKI (n = 49)	AKI-C = sCr>: 1.6 for 24–27 wks 1.1 for 28–29 wks 1.0 for 30–32 wks	– Renal volume lower in AKI (57 vs. 68, p = 0.04) – No difference in microalbuminuria, eGFR (sCr)	ELBWs with lower eGFR 99 vs. 107, p = 0.04
Maqsood et al. [85]	Born < 1000 g Age of F/U = 3 yrs AKI 2/3 (n = 23) AKI 1 (n = 87) No AKI (n = 112)	Modified KDIGO	– No difference in CKD (eGFR w/sCr) (4.2% vs. 4.1%)	ELBWs have growth restriction and higher DBP at discharge from NICU
Harer et al. [87]	Born < 1500 g Age of F/U = 5 yrs AKI (n = 20) No AKI (n = 14)	Modified KDIGO, sCR only	– AKI group with more renal dysfunction (65% vs. 14%, p = 0.01)	Renal dysfunction (HTN, low eGFR or proteinuria)

F/U follow-up, *yrs* years, *AKI* acute kidney injury, *ARF* acute renal failure; *sCr* serum creatinine, *UOP* urine output, *GFR* glomerular filtration rate, *ESRD* end stage renal disease, *AKI-C* acute kidney injury Bruel definition [62], *wks* weeks, *ELBWs* extremely low birth weight infants, *KDIGO* Kidney Disease Improving Global Outcomes, *DBP* diastolic blood pressure, *NICU* neonatal intensive care unit, *HTN* hypertension.

Reply: We have added these articles into our systematic review

Results:

- Page 9, line 42 “the smallest cohort size was 19…” In the text above, it states those with

less than 20 were excluded. How was this article included?

Reply: We agree with this comment and the article in question (Toffolo et al. 1997) has been removed.

- Nephrocalcinosis – I have a really hard time as including this as an outcome for long

term renal health of prematurity as your title implies. It is certainly an important short

term outcomes in the first few years – but ultimately most nephrocalcinosis resolves

after stopping diuretics for BPD and isn’t something that fits into the definition of CKD. I

would consider removing the papers about nephrocalcinosis only and removing this

section or changing the title of your paper.

Reply: We agree and have removed any papers that only investigated nephrocalcinosis and have removed any sections on it from our review. Toffolo et al. 1997 was the only article that only investigated nephrocalcinosis and was removed.

- GFR – This entire section is difficult to read and unfocused. The paragraphs should be rearranged to tell a story of the results that is logical and not be a list of what each study

found.

Reply: We have re-written the GFR paragraph in the results section to be more comprehensible and a better reflection of our findings.

o GFR: Where is the forest plot for this outcome?

- Blood pressure

o Where is the forest plot for this outcome?

Reply: We generated Forrest Plots for eGFR and BP as suggested using the RevMan software. However, we found significant variation in the study designs, outcome measures and many studies had to be excluded as the data was not available. Furthermore:

a. Different equations were used to calculate eGFR
b. A few studies provided only CysC levels and not the actual eGFR values
c. Varying study groups - some studies classified the participants into term, preterm small for gestational age and preterm normal for gestational age, whereas others didn't make such distinction or didn't have term control groups.
d. Outcomes measured at different age groups - both BP and eGFR changes with age.
e. One study provided only mean blood pressure measurement and not SBP and DBP

Hence, we were unable to include Forrest Plots in this revision as suggested.

There should be a paragraph describing the outcomes of proteinuria or albuminuria in

this population.

Reply: We have written a paragraph about proteinuria and albuminuria. It can be found on page 19. We have also discussed it in the discussion.

Discussion:

- Page 16 – line 20 "thus it is likely that in the short term, GFR is not significantly affected

by prematurity". This conclusion does not reflect the data presented in the article. It

should be modified.

Reply: We have modified this statement. It can now be found on Page 22 around line 20.

- Line 25, "this is likely the case…" I don't think we know why. The reduced nephron

number is one theory, but there are many contributing factors. To make a guess like this

is fine, but it should be framed as a hypothesis, not certainty.

Reply: We have altered this phrase to be framed as a hypothesis.

- Line 28-32 – this statement is out of place as your results didn't talk about CAD, obesity or metabolic disease, please remove both sentences.

Reply: We have removed these sentences.

- Lines 37-55 – this is an incorrect conclusion of your data. In your results you point out 4 studies that did find differences. Perhaps these are the best studies and methodology and the others had poor methodologies and incorrect findings – or not the right comparison groups, or too few of patients. This paragraph needs to be re-done.

Reply: This paragraph has been re-written.

- Page 17, nephrocalcinosis discussion – would remove this paragraph entirely.

Reply: This paragraph has been removed

Conclusion

- Line 42 – I would remove your age range and say childhood.
- Line 45 – your results do not support this statement and it should be removed.

Reply: These adjustments have been made.

Reviewer: 2

This is important work. I enjoyed reading and reviewing this. Would revise a few things as followed:
1. Conceptually you excluded two key papers with cohorts greater than 20 which observed long term renal outcomes after preterm birth. (South AM, Nixon PA, Chappell MC, et al. Renal function and blood pressure are altered in adolescents born preterm. Pediatr Nephrol. 2019;34(1):137-144. doi:10.1007/s00467-018-4050-z) (Sanderson KR, Chang E, Bjornstad E, et al. Albuminuria, Hypertension, and Reduced Kidney Volumes in Adolescents Born Extremely Premature. Front Pediatr. 2020;8:230. Published 2020 May 12. doi:10.3389/fped.2020.00230)

Reply: We thank the reviewer for their kind time, comments and suggestions. We will address these to improve our manuscript for resubmission. South et al. (2019) and Sanderson et al (2020) have been included in the study.

2. Following this point you state that you excluded cohorts where the n is less than 20 but in the second paragraph of the results, your very first sentence states that "Of the included studies, the smallest cohort size was 19.." this is contradictory to your exclusion criteria described in your methods and confusing to readers.

Reply: We agree with this comment and have removed Toffolo et al. (1997) for our included studies.

3. Also conceptually, you discuss renal dysfunction and hypertension in the intro as an explanation for your research question but nephrocalcinosis is under-developed from the very beginning. So it raises the question why did you evaluate nephrocalcinosis as an endpoint when it does not seem to be a part of the Brenner hypothesis or other prior research that you describe in the intro?

Reply: We agree with this point. On further discussion, we have decided to remove nephrocalcinosis as an outcome measure for our study. Toffolo et al. (1997) was the only study that investigated solely for nephrocalcinosis and thus it was removed for our included studies.

INTRODUCTION
1. The introduction is too long. I would recommend removing the entire 3rd paragraph as it moves into neurocognitive and metabolic disease outcomes which are not the focus of this paper.

Reply: We have removed paragraph 3 of the introduction.

2. Page 6 Line 13. Would remove "become" and replace with "is" as it will make the reader stumble. "As the ex-preterm population is living longer....understanding the effects of prematurity IS imperative in anticipating the likely chronic health outcomes...."
Reply: This has been corrected.

RESULTS
1. On page 11 line 24: Would correct as follows "Yael et al found that all 103 study participants with a history of very low birth weight and preterm birth had normal GFR values." Also you need to add at what age was this population studied as you did with other studies in this paragraph.

Reply: The sentence has been corrected and the age at outcome has been added.

2. On page 11 line 37: Would be consistent with representation of numbers. "Outcomes were measured at approximately 7.5 years, 8 years, 8.5 years..." And throughout the text would be consistent with representation of numbers (with the exception of a number at the beginning of a sentence which should always be spelled out). Its simple but can make things appear sloppy. **Reply:** We have made this change.

DISCUSSION
1. Ironically, this paragraph is started with a number 25 instead of the word "Twenty-five." Would start sentence with a number spelled out.

Reply: We have made this change.

VERSION 2 – REVIEW

REVIEWER	Matthew Harer University of Wisconsin-Madison School of Medicine and Public Health, Pediatrics
REVIEW RETURNED	25-May-2021

GENERAL COMMENTS	This revision is significantly improved and this will be a nice addition tot he literature. My biggest concern at this point is that the results section is challenging to read and would benefit from the addition of multiple figures or tables, even if forest plots are not possible. Perhaps a table for each outcome, with the left column those that favor that outcome and the right column those that do not favor the outcome with the studies with the highest 'weight' or 'number of patients' listed at the top? In the results, you quote the Harer study finding difference between preterm and term infants. That study actually compared VLBW infants with a history of AKI to those without a history of AKI, no term infants. A more correct statement would be "26% of the 34 subjects had an abnormal eGFR by cystatin C at 5 years of age." Askenazi is spelled wrong "Ashkenazi" page 63 line 13 and page 65 line 13

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Dr. Matthew Harer, University of Wisconsin-Madison School of Medicine and Public Health

Comments to the Author:

This revision is significantly improved and this will be a nice addition to the literature. My biggest concern at this point is that the results section is challenging to read and would benefit from the addition of multiple figures or tables, even if forest plots are not possible. Perhaps a table for each outcome, with the left column those that favor that outcome and the right column those that do not favor the outcome with the studies with the highest 'weight' or 'number of patients' listed at the top?

Reply: We kindly thank the reviewer for their time and suggestions.

We agree with this comment and have added two additional tables to the results section for our main outcomes (blood pressure and GFR). These tables demonstrate the studies that found a significant difference in outcomes between groups and compares them to those that did not. We have also listed the studies in order of the largest cohort size to the smallest as suggested. We are hopeful that these changes will make the revised manuscript easier to follow.

In the results, you quote the Harer study finding difference between preterm and term infants. That study actually compared VLBW infants with a history of AKI to those without a history of AKI, no term infants. A more correct statement would be "26% of the 34 subjects had an abnormal eGFR by cystatin C at 5 years of age."

Reply: We regret this error and have corrected it.

Askenazi is spelled wrong "Ashkenazi" page 63 line 13 and page 65 line 13

Reply: We regret this error and have corrected it.